

Optimal Talent Spending in Professional team Sports

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Sport Finance and Economics

1. Set up the profit-maximization problem

1.1 Winning technology

team i chooses talent spending x_i , while rival team j spends x_j .

We represent team i 's expected sporting strength as

$$w_i = \frac{x_i}{x_i + x_j}.$$

Here,

- x_i is team i 's talent spending,
- x_j is the rival's talent spending,
- $w_i \in (0, 1)$ is team i 's expected winning strength.

The key idea is that sporting strength is *relative*: team i 's success depends not only on its own talent, but also on the talent of its rival.

1.2 Revenue from winning

Let revenue depend on sporting success:

$$R_i(w_i) = m_i \left(aw_i - \frac{b}{2} w_i^2 \right).$$

where

- $m_i > 0$ is market size or revenue capacity,
- $a > 0$ is the initial revenue productivity of winning,
- $b > 0$ determines how quickly the revenue benefit of additional winning diminishes.

The negative quadratic term,

$$-\frac{b}{2}w_i^2,$$

introduces diminishing marginal revenue from winning.

1.3 Talent cost

Assume talent cost is linear:

$$C_i(x_i) = cx_i.$$

Therefore,

$$\frac{dC_i}{dx_i} = c.$$

Thus c is the constant marginal cost of talent.

1.4 Profit

Profit equals revenue minus talent cost:

$$\pi_i = R_i(w_i) - C_i(x_i).$$

Substituting the specific functions,

$$\pi_i = m_i \left(aw_i - \frac{b}{2}w_i^2 \right) - cx_i,$$

with

$$w_i = \frac{x_i}{x_i + x_j}.$$

The club solves

$$\max_{x_i} \pi_i.$$

2. Solve the maximization problem

2.1 How does talent affect winning?

Start from

$$w_i = \frac{x_i}{x_i + x_j}.$$

Differentiate with respect to x_i :

$$\frac{\partial w_i}{\partial x_i} = \frac{(x_i + x_j) - x_i}{(x_i + x_j)^2}.$$

Therefore,

$$\frac{\partial w_i}{\partial x_i} = \frac{x_j}{(x_i + x_j)^2} > 0.$$

So more talent increases team i 's expected sporting strength.

2.2 Marginal revenue from winning

Differentiate revenue with respect to winning:

$$R_i(w_i) = m_i \left(aw_i - \frac{b}{2} w_i^2 \right).$$

Hence,

$$\frac{\partial R_i}{\partial w_i} = m_i(a - bw_i).$$

This is the marginal revenue generated by an additional unit of sporting success.

2.3 Differentiate profit

Because talent affects revenue through winning, use the chain rule:

$$\frac{d\pi_i}{dx_i} = \frac{\partial R_i}{\partial w_i} \frac{\partial w_i}{\partial x_i} - \frac{dC_i}{dx_i}.$$

Substitute the three derivatives:

$$\frac{d\pi_i}{dx_i} = m_i(a - bw_i) \frac{x_j}{(x_i + x_j)^2} - c.$$

2.4 First-order condition

At an interior optimum,

$$\frac{d\pi_i}{dx_i} = 0.$$

Therefore,

$$m_i(a - bw_i) \frac{x_j}{(x_i + x_j)^2} - c = 0.$$

Rearranging,

$$m_i(a - bw_i) \frac{x_j}{(x_i + x_j)^2} = c.$$

The left-hand side is the marginal revenue generated by an additional unit of talent. The right-hand side is the marginal cost of talent.

Equivalently,

$$\frac{\partial R_i}{\partial w_i} \frac{\partial w_i}{\partial x_i} = c.$$

Since

$$\frac{dR_i}{dx_i} = \frac{\partial R_i}{\partial w_i} \frac{\partial w_i}{\partial x_i},$$

the general profit-maximization condition is

$$\frac{dR_i}{dx_i} = \frac{dC_i}{dx_i}.$$

Under the linear talent-cost assumption,

$$\frac{dR_i}{dx_i} = c.$$

Economic interpretation: the club acquires talent until the extra revenue generated by the next unit of talent equals the marginal cost of that talent.

3. Numerical calibration

Use

$$m_i = 100, \quad a = 2, \quad b = 1, \quad x_j = 10, \quad c = 2.$$

The first-order condition becomes

$$100(2 - w_i) \frac{10}{(x_i + 10)^2} = 2.$$

Because

$$w_i = \frac{x_i}{x_i + 10},$$

solve for x_i in terms of w_i :

$$w_i(x_i + 10) = x_i,$$

$$w_i x_i + 10w_i = x_i,$$

$$10w_i = x_i(1 - w_i),$$

so

$$x_i = \frac{10w_i}{1 - w_i}.$$

Then

$$x_i + 10 = \frac{10w_i}{1 - w_i} + 10 = \frac{10}{1 - w_i}.$$

Substitute this into the first-order condition:

$$100(2 - w_i) \frac{10}{\left(\frac{10}{1 - w_i}\right)^2} = 2.$$

Therefore,

$$10(2 - w_i)(1 - w_i)^2 = 2,$$

and

$$(2 - w_i)(1 - w_i)^2 = 0.2.$$

The economically relevant solution is approximately

$$w_i^* \approx 0.6194.$$

Then

$$x_i^* = \frac{10(0.6194)}{1 - 0.6194} \approx 16.27.$$

So, against a rival spending 10, the profit-maximizing club spends approximately 16.27 on talent and achieves expected sporting strength of approximately 61.9%.

4. Economic extension: change market size

Return to the first-order condition:

$$m_i(a - bw_i) \frac{x_j}{(x_i + x_j)^2} = c.$$

Suppose m_i increases.

For a given x_i , the marginal revenue generated by winning rises because the club has greater revenue capacity.

Therefore the marginal revenue product of talent rises.

The profit-maximizing response is to increase talent spending until the first-order condition is restored.

Thus, under the usual interior-solution conditions,

$$m_i \uparrow \implies x_i^* \uparrow \implies w_i^* \uparrow.$$

In words:

larger market $\rightarrow$ greater revenue capacity $\rightarrow$ more talent spending $\rightarrow$ higher expected wins.

This gives a microeconomic foundation for the link between market size, talent demand, and sporting performance.

5. Sloane extension: utility maximization

A club may value winning independently of profit.

Represent owner utility as

$$U_i = \pi_i + \lambda w_i, \quad \lambda > 0.$$

Here, λ captures the owner's additional utility from winning, prestige, championships, visibility, or legacy.

Differentiate with respect to x_i :

$$\frac{dU_i}{dx_i} = \frac{d\pi_i}{dx_i} + \lambda \frac{\partial w_i}{\partial x_i}.$$

Using the baseline derivative of profit,

$$\frac{dU_i}{dx_i} = m_i(a - bw_i) \frac{x_j}{(x_i + x_j)^2} - c + \lambda \frac{x_j}{(x_i + x_j)^2}.$$

Collect terms:

$$\frac{dU_i}{dx_i} = [m_i(a - bw_i) + \lambda] \frac{x_j}{(x_i + x_j)^2} - c.$$

The utility-maximizing first-order condition is therefore

$$[m_i(a - bw_i) + \lambda] \frac{x_j}{(x_i + x_j)^2} = c.$$

Compare the two cases.

For a profit-maximizing club,

$$m_i(a - bw_i) \frac{x_j}{(x_i + x_j)^2} = c.$$

For a utility-maximizing club,

$$[m_i(a - bw_i) + \lambda] \frac{x_j}{(x_i + x_j)^2} = c.$$

Since $\lambda > 0$, the marginal benefit of talent is higher for the utility-maximizing owner.

Under standard concavity conditions,

$$x_i^U > x_i^\pi.$$

Sloane's point: a utility- or win-maximizing club can rationally spend more on talent than a pure profit-maximizing club.

6. Why the revenue function has diminishing returns to winning

A richer revenue specification can include revenue earned independently of sporting performance:

$$R_i(w_i) = \bar{R}_i + m_i \left(\alpha w_i - \frac{\beta}{2} w_i^2 \right),$$

where

- $\bar{R}_i$ is baseline revenue earned regardless of sporting performance,
- m_i is the market-size or revenue-capacity parameter,
- $w_i \in [0, 1]$ is sporting strength,
- $\alpha > 0$ is the initial revenue productivity of winning,
- $\beta > 0$ determines how quickly the revenue benefit of additional winning diminishes.

The marginal revenue of winning is

$$\frac{\partial R_i}{\partial w_i} = m_i(\alpha - \beta w_i).$$

Differentiate again:

$$\frac{\partial^2 R_i}{\partial w_i^2} = -m_i\beta < 0.$$

Therefore $R_i(w_i)$ is strictly concave.

The economic meaning is straightforward: improving from a weak team to a competitive team may generate a large increase in attendance, sponsorship, media interest, and other revenue, while an additional increase in winning for an already dominant team may generate a smaller incremental revenue gain.

7. Diminishing returns from talent to winning

Recall

$$w_i = \frac{x_i}{x_i + x_j}.$$

The first derivative is

$$\frac{\partial w_i}{\partial x_i} = \frac{x_j}{(x_i + x_j)^2} > 0.$$

More talent increases sporting strength.

Differentiate again:

$$\frac{\partial^2 w_i}{\partial x_i^2} = -\frac{2x_j}{(x_i + x_j)^3} < 0.$$

Thus every additional unit of talent produces a progressively smaller increase in expected winning strength.

There are therefore two layers of diminishing returns:

Talent $\rightarrow$ Wins

and

Wins $\rightarrow$ Revenue.

8. Marginal revenue of talent

The full economic mechanism is

Talent $\rightarrow$ Wins $\rightarrow$ Revenue.

By the chain rule,

$$\frac{dR_i}{dx_i} = \frac{\partial R_i}{\partial w_i} \frac{\partial w_i}{\partial x_i}.$$

Using the baseline functions,

$$\frac{dR_i}{dx_i} = m_i(a - bw_i) \frac{x_j}{(x_i + x_j)^2}.$$

The first term,

$$m_i(a - bw_i),$$

is the additional revenue generated by more winning.

The second term,

$$\frac{x_j}{(x_i + x_j)^2},$$

is the additional winning generated by more talent.

Therefore,

$$MR_{\text{talent}} = MR_{\text{wins}} \times MP_{\text{talent}}.$$

At the profit-maximizing optimum,

$$MR_{\text{talent}} = MC_{\text{talent}}.$$

With constant marginal talent cost,

$$MR_{\text{talent}} = c.$$

9. Map of the argument

The complete logic can be read in the following sequence:

$$\max_{x_i} \pi_i$$

$\Downarrow$

$$w_i(x_i, x_j), \quad R_i(w_i), \quad C_i(x_i)$$

$\Downarrow$

$$\frac{d\pi_i}{dx_i} = 0$$

$\Downarrow$

$$\frac{\partial R_i}{\partial w_i} \frac{\partial w_i}{\partial x_i} = \frac{dC_i}{dx_i}$$

$\Downarrow$

$$MR_{\text{talent}} = MR_{\text{wins}} \times MP_{\text{talent}} = MC_{\text{talent}}.$$

From the baseline model we can then study three extensions:

1. change market size m_i ;
2. replace pure profit maximization with Sloane-style utility maximization;
3. examine diminishing returns in both the revenue function and the winning technology.